

SELECTION SEQUENCES FOR SELECTING THE BEST OF k NORMAL POPULATIONS

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1. Introduction

One way in which the problem of selecting the population among k normal populations with the largest mean (referred to as the best population) has been formulated, is to require a subset of the k populations, such that this subset contains the best population with given probability. This formulation is due to Gupta (1965) who shows how such subsets may be constructed for each fixed sample size and for the cases where the assumed common variance of the populations are known or unknown. Certain optimality properties are obtained also.

In the above mentioned formulation the size of the selected subset is random and could be large. This leads to another formulation of the selection problem where a subset of the k populations with *size at most a given number* is required and such that the best population is contained with given probability. (This is analogous to the problem of bounded length confidence intervals.) It is with this formulation that we shall concern ourselves in this paper.

Basic to our considerations is the concept of a *selection sequence*, which we will now define. Let for each $n \geq 1$ Π_n denote a subset of the k populations based on n observations from each population. Let π_1 be a $N(\mu_1, \sigma_1^2)$ population and assume without loss of generality that $\mu_1 \leq \dots \leq \mu_k$. Then we refer to π_k as the best population. We shall now say that $\{\Pi_n\}$ is a selection sequence if

$$P(\pi_k \in \Pi_n \text{ for all } n \geq 1) \geq P^* \text{ for all } \theta \in \theta$$

where P^* is given ($0 < P^* < 1$) and θ denotes an element of the parameter set θ . In this paper Π_n will depend on n observations from each population.

The idea of a selection sequence is analogous to that of a confidence sequence, introduced by Robbins (1970) and the advantages are similar.

Furthermore, in typical cases the sequence of subsets shrinks, so that eventually one population remains. A very important property of a selection sequence is as follows: in the process of sampling and computing members of the selection sequence one by one,

the experimenter may stop at any arbitrary random point, even depending on the observations, and the resulting subset contains the best population with probability at least P^* . In particular, the sampling may be terminated as soon as at most a given number of populations remains (if such a stage can be attained).

Let Π_n denote, for the moment, the subset provided by Gupta's procedure for the case where the k populations have a common known variance σ^2 . Then

$$\Pi_n = \{ \pi_s : \bar{X}_s(n) \geq \max_{1 \leq i \leq k} \bar{X}_i(n) - h\sigma/n^{1/2} \}$$

where $\bar{X}_s(n) = n^{-1} \sum_{j=1}^n \bar{X}_{sj}$ and $\bar{X}_{sj}$ denotes the j -th observation from π_s , $s = 1, \dots, k$; $j = 1, \dots, n$. The constant h is defined by the equation

$$\int_{-\infty}^{\infty} \Phi^{k-1}(x+h) d\Phi(x) = P^*$$

where Φ is the standard normal cdf. Then $P(\pi_k \in \Pi_n) \geq P^*$ for each $n \geq 1$. However, the sequence $\{\Pi_n\}$ does not define a selection sequence, as is easily seen by considering a parameter point with $\mu_k = \mu_{k-1}$ and using the Law of the Iterated Logarithm. The same is true for the sequence obtained from Gupta's procedure with σ unknown. This implies that, in order to find procedures for the second formulation above, based on selection sequences, we cannot simply use sequences of subsets provided by Gupta's procedures, but we have to find new sequences.

In the sections to follow, we shall construct selection sequences of structure similar to the sequence $\{\Pi_n\}$ just considered. We shall confine ourselves to the case where $\sigma_1^2, \dots, \sigma_k^2$ are assumed to be unknown and not necessarily equal. The case where a common, known variance is assumed may be treated similarly. In the literature the only attempted solution to the selection problem with unequal and unknown variances is given by Dudewicz and Dalal (1971) in an abstract. Their solution is a two-stage procedure and requires an indifference-zone formulation.

In a separate paper we intend to consider sequences with the property of eliminating inferior populations as the sampling proceeds and taking observations from the remaining populations at each stage only.

2. Construction of Selection Sequences

Using the notation of section 1 and neither assuming $\sigma_1^2, \dots, \sigma_k^2$ to be the same, nor to be known, we define for each $n \geq 2$

$$(1) \quad \Pi_n = \{ \pi_s : \bar{X}_s(n) \geq D_{ns} \} \quad \text{and} \quad \Pi_1 = \{ \pi_1, \dots, \pi_k \}$$

where, for each $j = 1, \dots, k$,

$$D_{nj} = \max_{1 \leq i \leq k} \bar{X}_i(n) - s_{nj}h(t, n)$$

and

$$s_{nj} = \max_{i \neq j} H(j, i, n) \text{ for } i = 1, \dots, k$$

with

$$H^2(j, i, n) = n^{-1} \sum_{\beta=1}^n \{(\bar{X}_{j\beta} - \bar{X}_{i\beta}) - (X_{j(n)} - X_{i(n)})\}^2$$

$$h(t, y) = [(ty)^{y-1} - 1]^{1/2}$$

$$t = (1 + a^2)^2/2$$

a is defined by the equation

$$(2) \quad \frac{1}{2} - (\arctan a)/\pi + a/(1+a^2)\pi = (1-P^*)/2(k-1).$$

In order to prove that $\{\Pi_n\}$ is a selection sequence, we need the following result of Robbins (1970, p. 1402).

Lemma 1

Let $\{X_n\}$ be a sequence of independent random variables, each with a $N(\mu, \sigma^2)$ distribution. Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$, $v_n^2 = n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$. Then $P(|\bar{X}_n - \mu| \geq$

$$v_n [(dn)^{n-1} - 1]^{1/2} \text{ for some } n \geq 2) \leq 1 - 2(\arctan x)/\pi + 2x/\pi(1+x^2) \quad (x > 0)$$

where $d = 1/4(1+x^2)^2$.

Theorem 1

The sequence defined in (1) is a selection sequence.

Proof:

$$P(\pi_k \notin \Pi_n \text{ for some } n \geq 1)$$

$$= P(\bar{X}_k(n) < D_{nk} \text{ for some } n \geq 2) \leq \sum_{i=1}^{k-1} P(X_k(n) < \bar{X}_i(n) - s_{nk}h(t, n) \text{ for some } n \geq 2)$$

$$\leq \sum_{i=1}^{k-1} P(X_i(n) - \bar{X}_k(n) - (\mu_i - \mu_k) \geq s_{nk} h(t, n) \text{ for some } n \geq 2)$$

$$\leq \sum_{i=1}^{k-1} P(X_i(n) - \bar{X}_k(n) - (\mu_i - \mu_k) \geq h(t, n) H(i, k, n) \text{ for some } n \geq 2)$$

$$\leq 1 - P^*$$

from lemma 1 and (2).

We may now define a second sequence based on $\{\Pi_n\}$ as follows:

$$(3) \quad \Pi'_n = \bigcap_{j=1}^n \Pi_j, \quad n \geq 1.$$

It follows immediately from theorem 1 that $\{\Pi'_n\}$ is also a selection sequence. Moreover, it is a *decreasing* sequence of subsets. Although it could happen that Π_n is empty for some n , the probability of such an event is at most $1 - P^*$ (since π_k is with probability $\geq P^*$ in each Π_n).

We shall now consider an interesting large sample property of the sequence $\{\Pi_n\}$. Let

$$S(n) = \sum_{j=1}^k I(\bar{X}_j(n) \geq D_{nj})$$

where $I(A)$ denotes the indicator function of the event A . Hence $S(n)$ is the number of populations in Π_n . Let us also denote by ℓ the number of indices j with $\mu_j = \mu_k$, $j = 1, \dots, k$.

Theorem 2

$S(n) \rightarrow \ell$ a.s. as $n \rightarrow \infty$ and moreover $\Pi_n = \{\pi_{k-\ell+1}, \dots, \pi_k\}$ a.s. for all n large.

Proof:

Denote by (Ω, Γ, P) the underlying probability space. In order to prove the theorem, it is sufficient to prove:

$$I(\bar{X}_j(n) \geq D_{nj}) \rightarrow 0 \text{ a.s. as } n \rightarrow \infty \text{ for } j = 1, \dots, k - \ell$$

$$I(\bar{X}_j(n) \geq D_{nj}) \rightarrow 1 \text{ a.s. as } n \rightarrow \infty \text{ for } j = k - \ell + 1, \dots, k.$$

Consider first the case where $j=1, \dots, k-\ell$. By the Strong Law of Large Numbers we obtain for each $\omega \in \Omega$ (ω not in a null set), for a $\delta > 0$ sufficiently small and for some $n_0(\omega)$

$$D_{nj} - \bar{X}_j(n) > \mu_k - \mu_j - \delta > 0$$

for $n \geq n_0(\omega)$.

On the other hand, if $j=k-\ell+1, \dots, k$ then $I(\bar{X}_j(n) \geq D_{nj})$

$$= I\left(\bigcap_{i=1}^k (\bar{X}_j(n) \geq \bar{X}_i(n) - s_{nj}h(t,n))\right)$$

$$= \prod_{i=1}^k I(\bar{X}_j(n) \geq \bar{X}_i(n) - s_{nj}h(t,n))$$

and each factor in this product tends to 1 a.s. as $n \rightarrow \infty$. For $i=1, \dots, k-\ell$ this follows in the same way as the first part of the theorem was proved. For $i=k-\ell+1, \dots, k$ we note that the Law of the Iterated Logarithm implies $(\bar{X}_j(n) - \bar{X}_i(n))/h(t,n) \rightarrow 0$ a.s. while s_{nj} tends to a positive constant.

3. Expected Number of Steps to obtain a Subset of Given Size

We now define the stopping variable N to be the first integer $n \geq 1$ such that $S(n) \leq m$ where m is an integer chosen in advance with $1 \leq m \leq k-1$. If $N < \infty$ the subset Π_N thus contains at most m populations, but contains the best population with probability at least P^* .

It follows immediately from theorem 2 that if $\ell \leq m$, then $N < \infty$ a.s. However, for the case $\ell \geq m+1$ it is clear that $N = \infty$ with positive probability. Since we feel that it will often be the case in practice that $\ell \leq m$, we shall now study the behaviour of N under this assumption. Since it seems unfeasible to obtain exact results, we consider asymptotic theory as $P^* \rightarrow 1$, which should be of importance, since usually P^* is close to 1 in practice. (Note that if $\ell \geq m+1$ the selection sequence may still be computed and stopped at any convenient point, the resulting subset having the desired probability of a correct selection).

Lemma 2

If $\ell \leq m$

(i) $N \uparrow \infty$ a.s. as $P^* \uparrow 1$

(ii) $\lim_{P^* \rightarrow 1} N / \log t = 1 / \log(1 + x_{(m+1)}^2)$

a.s. where $x_{(1)} \leq \dots \leq x_{(k)}$ are the ordered $x_1, \dots, x_k$ with $x_j = (\mu_k - \mu_j)/\gamma_j$ and $\gamma_j = \max_{i \neq j} (\sigma_i^2 + \sigma_j^2)^{1/2}$ for each $j=1, \dots, k$.

(iii) The set $\{N/\log t\}$ for $P^* < 1$ is uniformly integrable.

Proof:

(i) This follows immediately from the definition of N and the fact that

$$h(t, n) \uparrow \infty \text{ as } P^* \uparrow 1.$$

(ii) From the Strong Law of Large Numbers and (i) we obtain $\bar{X}_j(N) \rightarrow \mu_j$ a.s. as

$$P^* \rightarrow 1.$$

Thus for a.a. ω and for $\delta < \min_{i,j} (\mu_j - \mu_i)/3$ for all $i < j$ such that $\mu_i \neq \mu_j$ there exists a

$P_0^* = P_0^*(\omega, \delta)$ such that for $P^* \geq P_0^*$ we have $\mu_j - \delta < \bar{X}_j(N) < \mu_j + \delta$, $j = 1, \dots, k$ so that, if $\mu_i < \mu_j$, also $\bar{X}_i(N) < \bar{X}_j(N)$, $i, j = 1, \dots, k$.

From the definition of N we have $S(N) \leq m$

i.e. $\bar{X}_j(N) < D_{Nj}$ for at least $k-m$ j 's.

i.e. $\bar{X}_j(N) < \bar{X}_q(N) - s_{Nj}h(t, N)$ for $P^* \geq P_0^*$ for at least $k-m$ j 's and for some q such that $\mu_q = \mu_k$.

Since P_0^* may be chosen so large that $s_{Nj} > \gamma_j - \delta$, $j = 1, \dots, k$, this implies

$$x_j(\delta) > h(t, N) \text{ for at least } k-m \text{ } j\text{'s}$$

where $x_j(\delta) = (\mu_k - \mu_j + 2\delta)/(\gamma_j - \delta)$.

Hence $x_{(m+1)}(\delta) > h(t, N)$ and by the Strong Law of Large Numbers and the definition of h , we obtain

$$\liminf_{P^* \rightarrow 1} N/\log t \geq 1/\log(1 + x_{(m+1)}(\delta)) \text{ a.s.}$$

Since δ is arbitrary it may be taken to be 0 in the expression on the right-hand side.

Using the fact that $S(N-1) \geq m+1$, we get in the same way as above

$$\limsup_{P^* \rightarrow 1} N/\log t \leq 1/\log(1+x^2_{(m+1)}) \text{ a.s.}$$

(iii) Using a result of Bickel and Yahav (1968), it is sufficient to prove that

$$\sum_{n=1}^{\infty} P_0^* < P^* < 1 \sup P(N/\log t > n) < \infty \text{ for some } P_0^* < 1.$$

Let r be the largest integer less than or equal to $n \log t$. Then

$$\begin{aligned} P(N > r) &\leq P(S(r) \geq m+1) \leq P\left(\sum_{j=1}^{k-\ell} I(\bar{X}_j(r) \geq D_{rj}) \geq m-\ell+1\right) \\ &\leq \sum_{j=1}^{k-\ell} P(\bar{X}_j(r) \geq D_{rj})/m \quad (\text{Markov's inequality}). \end{aligned}$$

Let now $Y_j(r) = \bar{X}_j(r) - \bar{X}_k(r) - (\mu_j - \mu_k)$, then, using Markov's inequality and the c_r -inequality (Loève (1963), p. 155) we obtain for $j=1, \dots, k-\ell$

$$\begin{aligned} P(\bar{X}_j(r) \geq D_{rj}) &\leq P(\bar{X}_j(r) \geq \bar{X}_k(r) - s_{rj}h(t, r)) \\ &\leq E(Y_j(r) + s_{rj}h(t, r))^4 / (\mu_k - \mu_j)^4 \\ &\leq \{ 8E(Y_j(r))^4 + 8(h(t, r))^4 E(s_{rj}^4) \} / (\mu_k - \mu_j)^4 \\ &= O(n^{-1} \log n)^2 \text{ uniformly in } P^* > P_0^* \end{aligned}$$

$$\text{since } E(Y_j(r))^4 = O(r^{-2}), E(s_{rj}^4) = O(1)$$

and $r[h(t, r)]^2/\log(tr) \rightarrow 1$ uniformly in $P^* \in (P_0^*, 1)$ as $n \rightarrow \infty$ for some $P_0^* < 1$.

Theorem 3

If $\ell \leq m$

$$\lim_{P^* \rightarrow 1} EN/(-\log(1-P^*)) = 4/\log(1+x^2_{(m+1)})$$

with $x_{(m+1)}$ defined in lemma 2.

Proof:

This follows from lemma 2 and the fact that $\pi\alpha(1-P^*)/4(k-1) \rightarrow 1$ as $P^* \rightarrow 1$ (which may be derived from (2)).

We now turn to the stopping variable N' , defined to be the first integer $n \geq 1$ such that $S'(n) \leq m$, where $S'(n)$ is the number of populations in Π'_n . Since $S'(n) \leq S(n)$, it

follows immediately that $N' \leq N$, so that $N' < \infty$ a.s. if $\ell \leq m$. We also note that we may express N' as $N' = N_{(k-m)}$ where $N_{(1)} \leq \dots \leq N_{(k)}$ are the order statistics of $N_1, \dots, N_k$ and where N_i is the first integer $n \geq 2$ such that $\bar{X}_i(n) < D_{ni}$, $i=1, \dots, k$ (i.e. N_i is the number of steps needed to "eliminate" Π_i). Under the assumption $\ell \leq m$ it follows that $N_i < \infty$ a.s. for $i=1, \dots, k-\ell$. However, $N_i = \infty$ with positive probability for $i=k-\ell+1, \dots, k$. Since we may prove that for P^* large enough $N_i = \infty$ a.s. for $i=k-\ell+1, \dots, k$ it follows that we may restrict ourselves to N_i , $i=1, \dots, k-\ell$.

Lemma 3

(i) $N' \uparrow \infty$ a.s. as $P^* \uparrow 1$.

(ii) $\lim_{P^* \rightarrow 1} N'/\log t = 1/\log(1+x_{(m+1)}^2)$ a.s.

where $x_{(m+1)}$ is defined in lemma 2.

Proof:

(i) This follows since $N_i \uparrow \infty$ a.s. as $P^* \uparrow 1$ for $i=1, \dots, k-\ell$.

(ii) For $i=1, \dots, k-\ell$ we obtain

$$\bar{X}_i(N_i) < D_{N_i, i}$$

so that for P^* large enough

$$\bar{X}_i(N_i) < \bar{X}_q(N_i) - s_{N_i, i} h(t, N_i)$$

where $\mu_q = \mu_k$.

Hence

$$\liminf_{P^* \rightarrow 1} N_i/\log t \geq 1/\log(1+x_i^2) \text{ a.s.}$$

Using $\bar{X}_i(N_i-1) \geq D_{N_i-1, i}$ we obtain similarly

$$\limsup_{P^* \rightarrow 1} N_i/\log t \leq 1/\log(1+x_i^2) \text{ a.s.}$$

so that

$$\lim_{P^* \rightarrow 1} N_{(i)} / \log t = 1 / \log(1 + x_{(k-i+1)}^2) \text{ a.s.}$$

The result now follows from the fact that $N' = N_{(k-m)}$.

Theorem 4

If $\ell \leq m$

$$\lim_{P^* \rightarrow 1} EN' / (-\log(1 - P^*)) = 4 / \log(1 + x_{(m+1)}^2)$$

Proof:

Since $N' \leq N$, it follows from lemma 2(iii) that $N' / \log t$ is uniformly integrable. This, together with lemma 3, implies the theorem.

It is interesting to note that although N' is always bounded by N and will often be strictly less than N , the asymptotic expected sample sizes are equivalent. As a further point of comparison between $\{\Pi_n\}$ and $\{\Pi'_n\}$, we note that (see theorem 2) for n large Π_n consists of the ℓ "best" populations, while a similar property for Π'_n does not hold.

REMARK

Although the procedures proposed in this paper are for a different formulation of the selection problem than that of Gupta (see the Introduction) and thus not really comparable, some comparison is possible, as follows: assume first that the populations have a known, common variance σ^2 and let $k=2$. If $S(n)$ denotes the expected number of populations in the subset provided by Gupta, then

$$E(S(n)) = \Phi(h/2^{1/2} - (\mu_2 - \mu_1)n^{1/2}/2^{1/2}\sigma) + \Phi(h/2^{1/2} + (\mu_2 - \mu_1)n^{1/2}/2^{1/2}\sigma).$$

If we now require that $\lim_{P^* \rightarrow 1} E(S(n)) = 1$, then it follows that the smallest n that will make this possible satisfies

$$n \sim h^2 \sigma^2 / (\mu_2 - \mu_1)^2 \text{ if } \mu_2 > \mu_1.$$

Under the above assumption on the variances we construct the following selection sequence $\{\Pi_n\}$ analogous to the first considered in section 2: Let

$$\Pi_n = \{ \pi_s : \bar{X}_s(n) \geq \max_{1 \leq i \leq k} \bar{X}_i(n) - 2^{1/2} \sigma g(n, b) \}$$

where $g(n, b) = ((b^2 + \log n)/n)^{1/2}$ and where b satisfies

$$1 - \Phi(b) + b\phi(b) + \phi^2(b)/\Phi(b) = (1 - P^*)/(k-1).$$

That this constitutes a selection sequence follows from Robbins (1970), p. 1401.

Using methods similar to those of section 3 it is seen that if $k=2$ and N denotes the first n such that Π_n contains one population, then $EN \sim n$ as $P^* \rightarrow 1$.

If one considers Gupta's procedure for the case of a common, but unknown variance, it is reasonable to expect that the same asymptotic sample size would be obtained as above. (There are certain technical difficulties in the process to establish this.) If one specializes the procedures considered in this paper to this situation, one obtains $EN \sim h^2 / \log(1 + (\mu_2 - \mu_1)^2 / 2\sigma^2)$ so that

$EN/n \sim \psi / \log(1 + \psi/2)$ where $\psi = (\mu_2 - \mu_1)^2 / \sigma^2$ as $P^* \rightarrow 1$ and this is close to 2 for μ_2 close to μ_1 .

APPENDIX

For practical applications of the procedures in this paper, we give the following table of the constant a , which is defined in (2).

k	P*	0,75	0,80	0,85	0,90	0,95	0,99
2		4,960	6,260	9,409	12,680	25,439	127,319
3		10,120	12,680	16,937	25,439	50,917	254,646
4		15,235	19,064	25,439	38,180	76,386	381,972
5		20,339	25,439	33,933	50,917	101,853	509,298
6		25,439	31,810	42,426	63,652	127,319	636,623
7		30,536	38,180	50,917	76,386	152,785	763,950
8		35,632	44,548	59,407	89,119	178,250	891,276
9		40,727	50,917	67,896	101,853	203,716	1018,603
10		45,822	57,284	76,386	114,586	229,181	1145,930
11		50,917	63,652	84,875	127,319	254,646	1273,258
12		56,011	70,019	93,364	140,052	280,111	1400,584
13		61,105	76,386	101,853	152,785	305,576	1527,916
14		66,198	82,753	110,342	165,517	331,042	1655,242
15		71,292	89,119	118,830	178,250	356,507	1782,571

REFERENCES

BICKEL, P.J. and YAHAV, J.A. (1968). Asymptotically optimal Bayes and minimax procedures in sequential estimation. *Ann. Math. Statist.* **39**, 442-456.

- DUDEWICZ, E.J. and DALAL, S.R. (1971). Two-sample ranking procedures for means of normal populations with unknown variances (abstract). *Ann. Math. Statist.* **42**, 1486.
- GUPTA, S.S. (1965). On some multiple decision (selection and ranking) rules. *Technometrics* **7**, 225-246.
- LOËVE, M. (1963). *Probability Theory (Third Edition)*. D. van Nostrand Co. Princeton, N.J.
- ROBBINS, H. (1970). Statistical methods related to the law of the iterated logarithm. *Ann. Math. Statist.* **41**, 1397-1409.

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